

08.11.2016 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov – Dec 2016

Duration: 3 Hrs
Semester: V
Branch: Computer

Max. Marks: 100

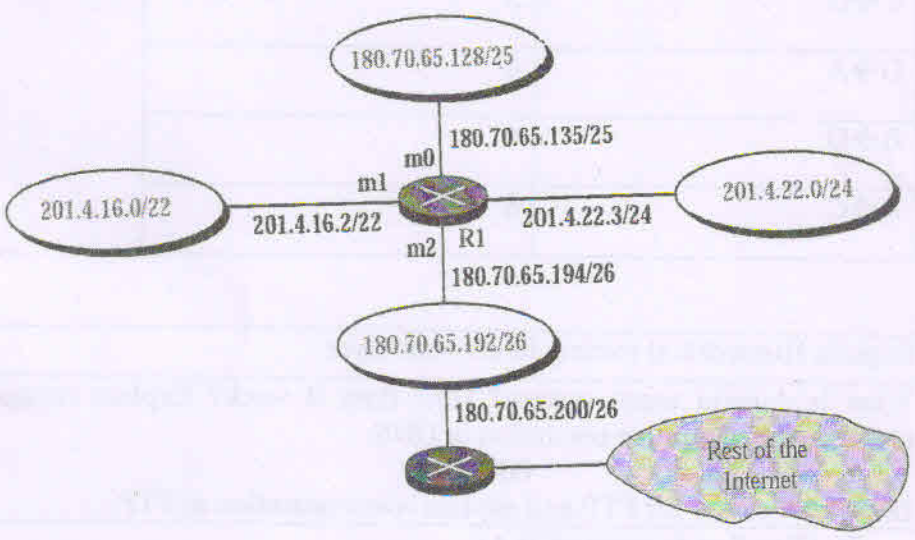
Class: TE

Name of the Course: Data Networks

Course Code: UCEC502

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams wherever necessary
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1	(i) What are the implications of mobility on TCP. (ii) Draw neat diagram and explain the scheme to solve hidden terminal problem.	(10)
Q2 (a)	Make a routing table for the router R1 using the configuration given in Figure below. Show the forwarding process if the packet arrives at R1 with the destination address 180.70.65.140.  OR An organization is granted the block 215.17.18.0/24. The administrator wants to create 8 subnets. a) Find the subnet mask. b) Find the number of addresses in each subnet. c) Find the first and last address in the first subnet. d) Find the first and last address in the last subnet. e) In case of classful addressing, which class does 215.017.18.2 belong to? Justify your answer.	(10)

Q2 (b)	(i) Compare IPV4 and IPV6 packet headers. (ii) What is subnetting and supernetting? How does their mask differ from the default mask? OR Explain the error reporting messages of ICMP.	(10)														
Q3 (a)	The following is the dump of the TCP header in the hexadecimal format. 05320017 00000001 00000000 500207FF 00000000 (i) What is the source port number? (ii) What is the destination port number? (iii)What is the length of the header? (iv)What is the window size? (v) What is the sequence number?	(10)														
Q3 (b)	Explain congestion policy in TCP. OR Explain the different timers used in TCP.	(10)														
Q4 (a)	For the given data, draw the network and determine the shortest path from source A to all destinations using Dijkstra's Algorithm. <table border="1"><thead><tr><th>Path</th><th>Cost</th></tr></thead><tbody><tr><td>A→B</td><td>2</td></tr><tr><td>B→C</td><td>3</td></tr><tr><td>C→D</td><td>5</td></tr><tr><td>D→A</td><td>4</td></tr><tr><td>B→D</td><td>6</td></tr><tr><td>A→C</td><td>8</td></tr></tbody></table>	Path	Cost	A→B	2	B→C	3	C→D	5	D→A	4	B→D	6	A→C	8	(10)
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Q4 (b)	Explain Hierarchical routing in network layer.	(10)														
Q5 (a)	What is domain name system? How does it work? Explain recursive resolution and iterative resolution in DNS. OR Draw neat model for FTP and explain communication in FTP.	(10)														
Q5 (b)	(i) Give five features of WAP (ii) What are the two types of messages in HTTP? Explain any one.	(10)														
Q6	(i) Compare UDP and TCP. (ii) Draw neat diagram and explain Snooping TCP.	(10)														
Q.6	OR What are MANETs? Give classification of routing protocols used in MANETS .Explain any one MANET routing protocol.	(10)														